

TECHNICAL MEMORANDUM

Crush Depth ROV: A Remotely Operated Vehicle for the MATE ROV World Championship

Competition seasons 2025 and 2026

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ABSTRACT. Crush Depth's ROV performs simulated environmental-research and industrial-inspection tasks for the MATE ROV competition. The structure is custom and FDM-printed from SolidWorks models; a Raspberry Pi 5 drives thrusters through ESCs and a PCA9685, and the C++ control layer is laid out for pilot ergonomics. This memorandum is written by the team's Co-CEO / CTO, who owned the electrical and software domains; the mechanical design in Section 1 is the work of the team's mechanical group. The vehicle competed at the World Championship in 2025 and 2026, and the author received the Engineering Presentation MVP award in 2026.

1 Mechanical Design

The vehicle's structure is custom, modeled in SolidWorks and printed FDM on Bambu Lab machines: an outer chassis, an internal housing that carries the electronics stack, and a TPU foam ballast printed to shape. A custom manipulator, designed against the competition's mission tasks, is the vehicle's working end.

Waterproofing is handled as a vehicle-level strategy rather than a per-part fix. Every penetration, housing, and connector has to hold at mission depth for the vehicle to score at all.

2 Electronics and Power

A Raspberry Pi 5 is the vehicle computer. It drives the thrusters through ESCs via a PCA9685 PWM driver, reads an MPU6050 IMU for attitude sensing, and carries the camera systems for the pilot. Power distribution runs through multiple buck converters stepping the tether supply down to the rails the electronics need.

Table 1. System summary.

Subsystem	Component	Notes
Vehicle computer	Raspberry Pi 5	control software in C++
Thruster drive	ESCs via PCA9685 PWM driver	—
Attitude	MPU6050 IMU	—
Vision	camera systems	pilot view
Power	buck converters from tether supply	multiple rails
Structure	FDM chassis, housing, TPU ballast	SolidWorks; Bambu Lab printers
End effector	custom manipulator	designed against mission tasks

3 Control Software

The control architecture, written in C++, is designed around pilot ergonomics: the layout of controls and behaviors is chosen for ease of piloting under mission time pressure, rather than for a clean mapping to actuators. Mission scores depend on what the pilot can actually execute.

4 Team and Role

Crush Depth is roughly 47 members of Clovis Community College's Engineering Renaissance Club, structured deliberately like a company per MATE guidelines. The author is Co-CEO / CTO, co-managing the team with Luke Salazar, and owns the electrical and software domains, promoted from Float Team Technical Lead. Operations at this scale are their own engineering problem: Gantt scheduling, BOM management, and holding deadlines across subteams.

At the 2026 World Championship the author won the **Engineering Presentation MVP** award.

5 Known Limitations

- The vehicle's total power budget was never formally characterized; converter sizing was validated by operation, not analysis.
- Manipulator payload capacity is unquantified.
- The hardware is currently not accessible to the author; this memorandum is limited to what was documented at build time.